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grupa 231

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Problema 5.15

Se considera urmatoarea metoda de rezolvare a ecuatiei f(x)=0:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_0)} \qquad .$$

Sa se determine ordinul de convergenta si eroarea asimtotica.

Fie r o radacina simpla atunci $r = x_n + e_n \; si \; r = x_0 + e_0$.

$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_0)} \rightarrow r - e_{n+1} = r - e_n - \frac{f(r-x_n)}{f'(r-x_0)} \rightarrow \; e_{n+1} = e_n + \frac{f(r-x_n)}{f'(r-x_0)}$ (1)

Folosind formula seriei Taylor : $f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a) + \cdots$

rescriem $f(x_n) \; si \; f'(x_0)$.

$$f(x_n) = f(r-e_n) = f(r) + f'(r)(x_n - r) + \frac{f''(r)}{2!}(x_n - r)^2 + \cdots = 0 + f'(r)(r - e_n - r) + \frac{f''(r)}{2!})(r - e_n - r)^2 + \cdots = -f'(r)e_n + \frac{{e_n}^2}{2}f''(r) + \cdots$$

$$f'(x_0) = f(r - e_0) = f'(r) + f''(r)(x_0 - r) + \frac{f'''(r)}{2!}(x_0 - r)^2 + \cdots \quad = f'(r) + \frac{f''(r)}{2!})(r - e_0 - r) \quad + \cdots = f'(r) - e_0 f''(r) + \cdots$$

Inlocuind in (1)

$$e_{n+1} = e_n - \frac{-f'(r)e_n + \frac{{e_n}^2}{2}f''(r)}{f'(r) - e_0 f''(r)} = \frac{e_n f'(r) - e_n e_0 f''(r) + f'(r)e_n - \frac{{e_n}^2}{2}f''(r)}{f'(r) - e_0 f''(r)}$$

$$= \frac{e_0 e_n - \frac{e_n^2}{2}f''(r)}{f'(r) - e_0 f''(r)} = \frac{2e_0 e_n - e_n^2 f''(r)}{2 * (f'(r) - e_0 f''(r))} \approx \frac{2e_0 e_n - e_n^2 f''(r)}{2f'(r)}$$

Daca $f''(r) \neq 0$ atunci ordinul de convergenta este 2.